

bug in dealing with data type at `one\_qubit\_decompose.py`

<https://github.com/Qiskit/qiskit/issues/9827>

ROW 72

bug in dealing with data type at one_qubit_decompose.py #9827

New issue

Closed

#9828



youngseok-kim1 opened on Mar 20, 2023

Environment

- Qiskit Terra version:

```
{'qiskit-terra': '0.23.0', 'qiskit-aer': '0.11.2', 'qiskit-ignis': '0.7.0', 'qiskit-ibmq-provider': '0.19.2', 'qiskit': '0.40.0', 'qiskit-nature': None, 'qiskit-finance': None, 'qiskit-optimization': None, 'qiskit-machine-learning': None}
```
- Python version:

```
python3.8
```
- Operating system: linux

What is happening?

I see error message as below:

```
"/home/labuser/.virtualenvs/controller/lib/python3.8/site-packages/qiskit/quantum_info/synthesis/one_qubit_decompose.py":10:
return self._params(unitary)
File "/home/labuser/.virtualenvs/controller/lib/python3.8/site-packages/qiskit/quantum_info/synthesis/one_qubit_decompose.py":10:
theta, phi, lam, phase = OneQubitEulerDecomposer._params_zyz(mat)
TypeError: argument 'unitary': type mismatch:
from=float64, to=complex128
```

this was with input

```
[[ 0.70710678  0.70710678]
 [ 0.70710678 -0.70710678]]
```

How can we reproduce the issue?

```
from qiskit.quantum_info import OneQubitEulerDecomposer

rzsxDecomposer = OneQubitEulerDecomposer('PSX')
M=[[ 0.70710678,  0.70710678], [ 0.70710678, -0.70710678]]
(theta, phi, lam, gamma) = rzsxDecomposer.angles_and_phase(M)
```

The error can be resolved by manually casting M to complex;

```
from qiskit.quantum_info import OneQubitEulerDecomposer

rzsxDecomposer = OneQubitEulerDecomposer('PSX')
M=[[ 0.70710678,  0.70710678], [ 0.70710678, -0.70710678]]
(theta, phi, lam, gamma) = rzsxDecomposer.angles_and_phase(np.array(M).astype(complex))
```

What should happen?

flag the data type error

Any suggestions?

add wrapper before pass it to rust so that every variable we are passing is casted to complex

Assignees

mtreinish

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix non-complex dtypes in OneQubitEulerDecomposer methods
Qiskit/qiskit

Notifications

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Participants



Fix non-complex dtypes in OneQubitEulerDecomposer methods #9828

<> Code

Merged jakelishman merged 3 commits into Qiskit:main from mtreinish:fix-float-angles on Mar 20, 2023

Conversation 8 Commits 3 Checks 0 Files changed 3

+50 -0



mtreinish commented on Mar 20, 2023

Member

Summary

This commit fixes a regression introduced in the rewrite of the internals of the OneQubitEulerDecomposer class to be in rust. Previously the angles() and angles_and_phase() methods of OneQubitEulerDecomposer would work with a non-complex dtype in the input, however because the rust component of the module is strictly typed to only work with a complex128 dtype passing the array to rust to compute the angles and phase of the unitary would fail. To address this limitation this commit casts the input matrix to be complex before passing it to the rust function.

Details and comments

Fixes #9827



Fix non-complex dtypes in OneQubitEulerDecomposer methods Verified 3e8b19e

mtreinish added stable backport potential Changelog: Bugfix mod: quantum info synthesis labels on Mar 20, 2023

mtreinish requested review from a team, alexanderivrii, ShellyGarion and ikkoham as code owners 2 years ago



qiskit-bot commented on Mar 20, 2023

Collaborator

Thank you for opening a new pull request.

Before your PR can be merged it will first need to pass continuous integration tests and be reviewed. Sometimes the review process can be slow, so please be patient.

While you're waiting, please feel free to review other open PRs. While only a subset of people are authorized to approve pull requests for merging, everyone is encouraged to review open pull requests. Doing reviews helps reduce the burden on the core team and helps make the project's code better for everyone.

One or more of the the following people are requested to review this:

- @Qiskit/terra-core
- @ikkoham



jakelishman reviewed on Mar 20, 2023

View reviewed changes

jakelishman left a comment

Member

Just one comment about the release note, but the code change looks good.



releasenotes/notes/fix-type-angles-euler-decompose-233e5cee7205ed03.yaml Outdated Show resolved

Reviewers

jakelishman	✓
alexanderivrii	🛡️
ShellyGarion	🛡️
ikkoham	

Assignees

No one assigned

Labels

Changelog: Bugfix mod: quantum info
stable backport potential synthesis

Projects

None yet

Milestone

0.23.3

Development

Successfully merging this pull request may close these issues.

bug in dealing with data type at one_qubit_...

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4 participants

